MIDHUNRAJ

DATA ENGINEER

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LinkedIn | GitHub

PROFESSIONAL SUMMARY

Data engineering professional transitioning from a Computer Science academic background, with hands-on experience building end-to-end ETL pipelines using Python, PySpark, Apache Airflow, Docker, PostgreSQL, and AWS S3. Experienced in Bronze-Silver-Gold architecture, star schema data warehousing, data profiling, cleaning, and workflow orchestration. Strong foundation in Python, SQL, and distributed data processing.

TECHNICAL SKILLS

- Languages: Python, SQL
- Big Data Processing: Apache Spark, PySpark, Spark Core, DataFrame API, Spark SQL
- Workflow Orchestration: Apache Airflow
- Cloud & Storage: AWS S3, boto3
- Databases & Warehousing: PostgreSQL, SQLite, Star Schema, Dimensional Modeling
- APIs: REST APIs, NewsAPI
- Data Quality: Data Profiling, Data Validation, Data Cleaning
- Containerization: Docker, Docker Compose
- Data Architecture: Medallion Architecture (Bronze / Silver / Gold), ETL pipeline design
- Monitoring & Logging: Structured logging
- Tools: Git, GitHub, pandas, YAML configuration management
- Additional: HTML, CSS, Django, Flask

PROJECTS

Retail Sales ETL Pipeline

PySpark, Apache Airflow, Docker, PostgreSQL, SQLite | github.com/midhunraj-303/retail-sales-etl-pyspark-airflow

- Built an end-to-end batch ETL pipeline using PySpark to process 4,310 retail sales records with a Bronze-Silver-Gold medallion architecture
- Developed data profiling and cleaning stages covering missing value analysis, duplicate detection, data validation, and text standardization using PySpark DataFrame operations
- Implemented business transformations including date features, customer age grouping, order-size classification, and shipping-type analysis
- Orchestrated the pipeline with Apache Airflow DAGs, including retry logic and execution timeouts, and containerized the environment using Docker
- Loaded processed data into PostgreSQL and SQLite, with additional Parquet and CSV exports for downstream analytics

News API ETL Pipeline

PySpark, PostgreSQL, Apache Airflow, Docker | github.com/midhunraj-303/news-api-etl-pipeline

- Built an end-to-end batch ETL pipeline to ingest live news data from NewsAPI and process it using PySpark
- Implemented a Bronze-Silver-Gold medallion architecture with dedicated data profiling, cleaning, transformation, and aggregation stages
- Designed a star schema data warehouse in PostgreSQL centered on a fact_articles table with source, author, and date dimension tables
- Orchestrated the pipeline using Apache Airflow with configurable retry logic and containerized the environment using Docker and Docker Compose
- Managed API secrets and configuration using environment variables and YAML configuration across local and Airflow execution environments

PROFESSIONAL EXPERIENCE

Assistant Professor, Computer Science *Wayanad, Kerala*

Model College Meenangadi | 2019 - 2021, 2022 - 2026

- Delivered lectures and laboratory sessions in programming, data structures, databases, and core Computer Science subjects
- Designed course materials, assignments, and assessments, and evaluated student performance
- Guided and mentored student projects, developing structured problem-solving, technical communication, and project-planning skills

EDUCATION

M.Sc. Computer Science *2017*

University of Calicut

B.Sc. Computer Science *2015*

University of Calicut

UGC NET - Computer Science and Applications *2019*

CERTIFICATIONS

- Complete Data Engineering Bootcamp: SQL, ETL & Data Pipeline - Udemy (43 hours, 2026)
- Learn Python Programming Masterclass - Udemy (61.5 hours, 2026)
- Python Masterclass 2026: Build 24 Real World Python Projects - Udemy (80.5 hours, 2026)